ANSWER KEY & MARKING SCHEME SET A

Chapters 9 & 10: **Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 4	2 diagonals and 2 mid-lines
2	**(a)** no line of symmetry	no equal sides
3	**(d)** E	top matches bottom
4	**(c)** looks the same after rotating less than a full turn about a point	definition
5	**(b)** 90°	360 ÷ 4
6	**(c)** to the left of zero	position
7	**(a)** 0	cancel
8	**(a)** −3	closer to zero
9	**(d)** −5	$3 - 8$
10	**(c)** −4	move left

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The fixed point about which the figure is turned.	1+1
12	No line symmetry; has rotational symmetry of order 2.	1+1
13	−20 and +40.	1+1
14	−11 and −9.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	No two sides are equal so no fold can match halves (3).	1+2
16	350 m (3).	1+2
17	0 and 8 look the same (2); 2 does not (1).	2+1
	OR: An isosceles triangle (3).	1+2
18	−3, −2, −1, 0, 1, 2; sum −3 (3).	1+2
	OR: $8 - 15 = -7$°C (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Every diameter is an axis (2). (ii) Infinite (1). (iii) Polygons have limited lines (2).	2+1+2
	OR: (i) Line symmetry (1). (ii) One (1). (iii) No (1). (iv) Butterfly (2).	1+1+1+2
20	(i) Distance from zero (1). (ii) 12, 9, 0 (2). (iii) 21 (2).	1+2+2
	OR: (i) −25, +8, −12 (1). (ii) −29 m (2). (iii) 29 m (1). (iv) 29 below zero (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapters 9 & 10: **Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1	no rotational symmetry
2	**(b)** 24	number of spokes
3	**(c)** rangoli and architecture	art
4	**(d)** 5	equal to sides
5	**(b)** 45°	360 ÷ 8
6	**(c)** 0	neutral
7	**(c)** 3°C	$-4+7$
8	**(c)** 5	add integers using signs
9	**(c)** 0	add integers using signs
10	**(a)** −5	add integers using signs

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Each point and its image are at equal distances from the mirror line on opposite sides.	1+1
12	A circle; every diameter is a line of symmetry.	1+1
13	Both are 15 (distance from zero).	1+1
14	10 units.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fold paper and cut shapes at the fold (1.5); unfold to see symmetric design (1.5).	1½+1½
16	−200 + 75 − 40 = −165 m (3).	1+2
17	Rectangle and rhombus (3).	1+2
	OR: One, along the diagonal joining the meeting points of equal sides (3).	1+2
18	Start at −2, move 5 steps right, reach 3 (3).	1+2
	OR: Opposites cancel; e.g. −6 + 6 = 0 (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Correct completion (2). (ii) Line symmetry (1). (iii) Leaf or face (2).	2+1+2
	OR: (i) Triangle 3, square 4, pentagon 5 (3). (ii) 5 × 72° = 360° (2).	3+2
20	(i) Arrows on the line (3). (ii) 3 and −11 (2).	3+2
	OR: (i) $a-b=a+(-b)$ with an example (3). (ii) 8 and −8 (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 9 & 10: **Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 2	two mid-lines
2	**(b)** 1 line of symmetry	equal sides
3	**(d)** H	two lines
4	**(d)** centre of rotation	fixed point
5	**(c)** 120°	360 ÷ 3
6	**(b)** −5	additive inverse
7	**(d)** −1	negative
8	**(a)** −8	farthest left
9	**(c)** −8	add the negatives
10	**(b)** 2	$-3+5$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	2.	1+1
12	72°.	1+1
13	No; it is neither positive nor negative.	1+1
14	5°C.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Number of times a figure fits on itself in a full turn; square has 4 (3).	1+2
16	13 (3).	1+2
17	Hexagon (6 > 5), as it fits 6 times in a full turn (3).	1+2
	OR: Line symmetry involves folding (1.5); rotational involves turning about a point (1.5).	1½+1½
18	Negative; since −12 has larger absolute value (3).	1+2
	OR: $(-5)+(-3)=-8$, a debt of 8 (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 5 (1). (ii) 72° (1). (iii) Yes (1). (iv) Correct drawing (2).	1+1+1+2
	OR: (i) Top half equals bottom half (3). (ii) B, C, D (2).	3+2
20	(i) −400, +8848 (1). (ii) 9,248 m (2). (iii) No (1). (iv) 4,224 m (1).	1+2+1+1
	OR: (i) 4; −7; −6 (3). (ii) Using inverse (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

Chapters 9 & 10: **Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Rectangle	half-turn matches
2	**(d)** 1 line of symmetry	main diagonal
3	**(b)** stays the same	congruent
4	**(d)** 5	equal to sides
5	**(b)** 7	equal to the number of sides
6	**(b)** −3	below ground
7	**(b)** −4°C	$2 - 6$
8	**(c)** −10	add integers using signs
9	**(a)** −3	add integers using signs
10	**(b)** −6	add integers using signs

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Balance and beauty in buildings such as temples and the Taj Mahal.	1+1
12	No (order 1).	1+1
13	−7 m.	1+1
14	−2.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Both have vertical and horizontal lines of symmetry (3).	1+2
16	55 m (3).	1+2
17	Five lines from each vertex to the midpoint of the opposite side (3).	1+2
	OR: MOM, WOW and TOT (1.5); each letter has a vertical line of symmetry (1.5), but B does not.	1½+1½
18	3 steps west, −3 (3).	1+2
	OR: Start with 4 red; remove 6 red by adding 2 pairs; left with 2 yellow = 2 (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A line of symmetry is a mirror line (3). (ii) Fan, wheel (2).	3+2
	OR: (i) n lines, order n and angle 360/n (3). (ii) 8 lines, order 8, 45° (2).	3+2
20	(i) Brahmagupta gave rules (Brahmasphutasiddhanta) for debts and fortunes (3). (ii) To represent debts and below-zero quantities (2).	3+2
	OR: (i) Move left 5 from 3 to −2 (2). (ii) 3 yellow, 5 red leaves 2 red (2). (iii) Suitable story (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.